CYBERSRS

Network Configuration Management System

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Design a secure network configuration management system for a university that manages configuration changes across routers, switches, wireless controllers, and other network infrastructure. Authorized network administrators should be able to review current configurations, submit configuration changes, track change history, and restore previously approved configurations when necessary. The system should detect unauthorized or unexpected configuration changes, generate security alerts, maintain audit logs of configuration activity and administrator actions, and restrict administrative functions based on approved roles. The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder source statement E-001

Detail: Design a secure network configuration management system for a university that manages configuration changes across routers, switches, wireless controllers, and other network infrastructure.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: Authorized network administrators should be able to review current configurations, submit configuration changes, track change history, and restore previously approved configurations when necessary.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: The system should detect unauthorized or unexpected configuration changes, generate security alerts, maintain audit logs of configuration activity and administrator actions, and restrict administrative functions based on approved roles.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Status: confirmed

Evidence ids: E-004

2. Functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

FR-001 - manage configuration changes across routers, switches, wireless controllers, and other network infrastructure

Statement: The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure.

Rationale: Source obligation E-001_O01.

Priority: must

Dimensions: business

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: the system manages configuration changes across routers, switches, wireless controllers, and other network infrastructure

When: the system initiates a configuration change operation

Then: the resulting configuration is applied without error to the targeted device

Method: test

Observable evidence: External test capture of the resulting response

FR-002 - allow Authorized network administrators to review current configurations

Statement: The system shall allow Authorized network administrators to review current configurations.

Rationale: Source obligation E-002_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized network administrators review current configurations

When: initiate the review of current configurations

Then: the requested configuration items are returned for examination

Method: inspection

Observable evidence: External test capture of the resulting response

FR-003 - allow Authorized network administrators to submit configuration changes

Statement: The system shall allow Authorized network administrators to submit configuration changes.

Rationale: Source obligation E-002_O02.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized network administrators submit a configuration change

When: initiate submission of a configuration change on the named object

Then: the object's resulting policy, job, submitted change, or history is observed as updated

Method: analysis

Observable evidence: External test capture of the resulting state

FR-004 - allow Authorized network administrators to track change history

Statement: The system shall allow Authorized network administrators to track change history.

Rationale: Source obligation E-002_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized network administrators track change history

When: initiate tracking of change history on the named object

Then: the object's resulting policy, job, submitted change, or history is observed as accessible and accurate

Method: demonstration

Observable evidence: External test capture of the inspection record

FR-005 - allow Authorized network administrators to restore previously approved configurations when necessary

Statement: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.

Rationale: Source obligation E-002_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: Authorized network administrators restore previously approved configurations

When: initiate restoration of a previously approved configuration on the named object

Then: the object's resulting policy, job, submitted change, or restored state is observed as valid and consistent with approval records

Method: test

Observable evidence: External test capture of the resulting state

FR-006 - detect unauthorized or unexpected configuration changes

Statement: The system shall detect unauthorized or unexpected configuration changes.

Rationale: Source obligation E-003_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: the system detects an unauthorized or unexpected configuration change

When: the system initiates the detection process

Then: an alert is generated and logged in the audit trail

Method: test

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-51ce9ae15f38 is TBD.

FR-007 - generate security alerts

Statement: The system shall generate security alerts.

Rationale: Source obligation E-003_O02.

Priority: should

Dimensions: business; security

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: a security alert condition is triggered by an anomalous configuration event

When: the system evaluates and responds to the alert condition

Then: the alert exists in the designated monitoring channel as captured externally

Method: inspection

Observable evidence: External test capture of the resulting state

FR-008 - maintain audit logs of configuration activity and administrator actions

Statement: The system shall maintain audit logs of configuration activity and administrator actions.

Rationale: Source obligation E-003_O03.

Priority: should

Dimensions: business; auditability; data

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: an administrator modifies a network configuration

When: the system records the action and all associated metadata

Then: The system records the action and all associated metadata.

Method: analysis

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-3aae42e94793 is TBD.; Decision Q-64ef2e28335c is TBD.

FR-009 - restrict administrative functions based on approved roles

Statement: The system shall restrict administrative functions based on approved roles.

Rationale: Source obligation E-003_O04.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: an administrator attempts to perform a restricted administrative function

When: the system evaluates the role-based access policy

Then: the request is denied if the role lacks approved permissions

Method: demonstration

Observable evidence: External test capture of the resulting state

FR-010 - provide a centralized dashboard

Statement: The system shall provide a centralized dashboard.

Rationale: Source obligation E-004_O01.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: system provides a centralized dashboard

When: the system initiates the display of network configuration data

Then: The system initiates the display of network configuration data and the external test capture records the resulting response.

Method: demonstration

Observable evidence: External test capture of the resulting response

FR-011 - integrate with existing university identity services

Statement: The system shall integrate with existing university identity services.

Rationale: Source obligation E-004_O02.

Priority: should

Dimensions: business; integration

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: system integrates with existing university identity services

When: the system performs a login authentication request via university identity protocols

Then: an authenticated session is established and persisted across operations

Method: test

Observable evidence: External test capture of the resulting state

FR-012 - support multiple campus locations

Statement: The system shall support multiple campus locations.

Rationale: Source obligation E-004_O03.

Priority: should

Dimensions: business

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: system supports multiple campus locations

When: the system is initiated from each stated campus location

Then: use succeeds at all locations with consistent configuration access

Method: test

Observable evidence: External test capture of the inspection record

3. Non-functional requirements

Applicability reason: Independent source obligations; unspecified choices remain TBD.

Requirements

NFR-001 - be secure

Statement: The system shall be secure.

Rationale: Source obligation E-001_O02.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-001

Acceptance criteria

Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data exfiltration via network interfaces.

When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD.

Then: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or response metrics are stated in the draft plan.

Method: inspection

Observable evidence: External test capture of the inspection record

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-002 - remain secure

Statement: The system shall remain secure.

Rationale: Source obligation E-004_O04.

Priority: should

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain secure under all operational conditions.

When: An analyst reviews the security assurance plan and evaluates whether it includes measurable targets for threat detection, response time, and configuration drift

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-7b2e5f89a0b1 is TBD.

NFR-003 - remain reliable

Statement: The system shall remain reliable.

Rationale: Source obligation E-004_O05.

Priority: should

Dimensions: availability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain reliable under all operational conditions.

When: An analyst reviews the reliability assurance plan and evaluates whether it includes measurable targets for failure detection, recovery time, and service uptime

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-819bb986f419 is TBD.

NFR-004 - remain scalable

Statement: The system shall remain scalable.

Rationale: Source obligation E-004_O06.

Priority: should

Dimensions: performance

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain scalable under all operational conditions.

When: An analyst reviews the scalability assurance plan and evaluates whether it includes measurable targets for throughput, latency, and resource utilization growth

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-543394346815 is TBD.

NFR-005 - remain manageable

Statement: The system shall remain manageable.

Rationale: Source obligation E-004_O07.

Priority: should

Dimensions: usability

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: The system shall remain manageable under all operational conditions.

When: An analyst reviews the manageability assurance plan and evaluates whether it includes measurable targets for configuration change frequency, audit coverage, and

Then: The plan's measurable targets remain TBD or unconfirmed

Method: inspection

Observable evidence: External test capture of the resulting response

Unresolved decisions: Decision Q-5268b96e41b0 is TBD.

Related requirement ids: FR-001; FR-010; FR-011; FR-012

4. Security, privacy and compliance

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-008

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005

7. Compliance and policy applicability matrix

Open decisions: Decision Q-51ce9ae15f38 is TBD.; Decision Q-3aae42e94793 is TBD.

8. Architecture

Summary: Proposed logical components and interactions mapped to canonical source obligations.

Items

Actor: Authorized network administrators

Detail: Proposed logical endpoint for the source-stated actor: Authorized network administrators.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

Component name: Actor: Authorized network administrators

Responsibility: Represent the source-stated actor: Authorized network administrators.

Configuration Management Engine

Detail: Proposed logical owner for FR-001, FR-002, FR-003, FR-004, FR-005, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-002; E-004

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Configuration Management Engine

Responsibility: The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure. The system shall allow Authorized network administrators to review current configurations. The system shall allow Authorized network administrators to submit configuration changes. The system shall allow Authorized network administrators to track change history. The system shall allow Authorized network administrators to restore previously approved configurations when necessary. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Security Monitoring and Alerting Module

Detail: Proposed logical owner for FR-006, FR-007, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-006; FR-007; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Security Monitoring and Alerting Module

Responsibility: The system shall detect unauthorized or unexpected configuration changes. The system shall generate security alerts. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Audit Logging and Compliance Engine

Detail: Proposed logical owner for FR-008, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-008; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Audit Logging and Compliance Engine

Responsibility: The system shall maintain audit logs of configuration activity and administrator actions. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Role-Based Access Control Service

Detail: Proposed logical owner for FR-009, NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-003; E-004

Requirement ids: FR-009; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Component name: Role-Based Access Control Service

Responsibility: The system shall restrict administrative functions based on approved roles. The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Centralized Dashboard Interface

Detail: Proposed logical owner for FR-010.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-010

Component name: Centralized Dashboard Interface

Responsibility: The system shall provide a centralized dashboard.

University Identity Integration Module

Detail: Proposed logical owner for FR-011.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-011

Component name: University Identity Integration Module

Responsibility: The system shall integrate with existing university identity services.

Multi-Campus Deployment Manager

Detail: Proposed logical owner for FR-012.

Status: proposed

Evidence ids: E-004

Requirement ids: FR-012

Component name: Multi-Campus Deployment Manager

Responsibility: The system shall support multiple campus locations.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-002, FR-003, FR-004, FR-005.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

From component: Actor: Authorized network administrators

To component: Configuration Management Engine

Data or action: The system shall allow Authorized network administrators to review current configurations. The system shall allow Authorized network administrators to submit configuration changes. The system shall allow Authorized network administrators to track change history. The system shall allow Authorized network administrators to restore previously approved configurations when necessary.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Configuration Management Engine

To component: Security Monitoring and Alerting Module

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Configuration Management Engine

To component: Audit Logging and Compliance Engine

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Configuration Management Engine

To component: Role-Based Access Control Service

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Monitoring and Alerting Module

To component: Configuration Management Engine

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Monitoring and Alerting Module

To component: Audit Logging and Compliance Engine

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Security Monitoring and Alerting Module

To component: Role-Based Access Control Service

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Audit Logging and Compliance Engine

To component: Configuration Management Engine

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Audit Logging and Compliance Engine

To component: Security Monitoring and Alerting Module

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Audit Logging and Compliance Engine

To component: Role-Based Access Control Service

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Role-Based Access Control Service

To component: Configuration Management Engine

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Role-Based Access Control Service

To component: Security Monitoring and Alerting Module

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Proposed source-defined interaction

Detail: Proposed directed path for NFR-001, NFR-002, NFR-003, NFR-004, NFR-005.

Status: proposed

Evidence ids: E-001; E-004

Requirement ids: NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

From component: Role-Based Access Control Service

To component: Audit Logging and Compliance Engine

Data or action: The system shall be secure. The system shall remain secure. The system shall remain reliable. The system shall remain scalable. The system shall remain manageable.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002, FR-003, FR-004, FR-005.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-002; FR-003; FR-004; FR-005

Untrusted component: Actor: Authorized network administrators

Trusted component: Configuration Management Engine

Enforcement: The system shall allow Authorized network administrators to review current configurations. The system shall allow Authorized network administrators to submit configuration changes. The system shall allow Authorized network administrators to track change history. The system shall allow Authorized network administrators to restore previously approved configurations when necessary.

9. Threats

Threats

Id: THR-001

Asset: User identities and authenticated access

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Impact: Unauthorized use of a legitimate identity could expose resources or alter system state.

Control requirement ids: FR-010; FR-011; FR-012; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-002

Asset: Role and permission boundaries

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Impact: A lower-privilege user could access restricted functions or sensitive resources.

Control requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009

Evidence ids: E-002; E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-003

Asset: Integration trust paths

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Impact: Untrusted integration data could cause incorrect decisions or hide security events.

Control requirement ids: FR-010; FR-011; FR-012; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-004

Asset: Security and administrative audit evidence

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Impact: Investigators could lose reliable evidence of the events required by the brief.

Control requirement ids: FR-006; FR-007; FR-008; FR-009

Evidence ids: E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Id: THR-005

Asset: Monitoring and security alerts

Actor: Hypothetical attacker, privileged insider or failure cause

Scenario: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Impact: Relevant suspicious activity could remain undetected or receive delayed investigation.

Control requirement ids: FR-006; FR-007; FR-008; FR-009

Evidence ids: E-003

Residual risk: Unverified planning hypothesis. Linked requirements define intended behavior, not proven prevention. Validate enforcement and residual exposure during design and acceptance review; likelihood and severity are not established.

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

Risks

Id: RISK-001

Description: Derived risk affecting User identities and authenticated access: Unauthorized use of a legitimate identity could expose resources or alter system state.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker uses a compromised identity or authenticated session to impersonate an authorized user and exercise that user's permitted operations.

Mitigation: Review impersonation and compromised-identity paths against the stated access duties; confirm any additional identity or session protections with the stakeholder. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-010; FR-011; FR-012; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-002

Description: Derived risk affecting Role and permission boundaries: A lower-privilege user could access restricted functions or sensitive resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A principal exploits an incorrect role assignment or permission check to perform operations outside the source-defined authorization boundary.

Mitigation: Test permitted and denied role/resource pairs against the linked authorization duties; review ambiguous permission assignments with the resource owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009

Evidence ids: E-002; E-003

Owner: TBD

Id: RISK-003

Description: Derived risk affecting Integration trust paths: Untrusted integration data could cause incorrect decisions or hide security events.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: A compromised connected system or integration path supplies forged, altered or missing identity, event or policy data used by the source-defined integration.

Mitigation: Review trust assumptions at the specified integration boundaries with missing or altered input fixtures; confirm the required failure handling with the owner. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-010; FR-011; FR-012; NFR-002; NFR-003; NFR-004; NFR-005

Evidence ids: E-004

Owner: TBD

Id: RISK-004

Description: Derived risk affecting Security and administrative audit evidence: Investigators could lose reliable evidence of the events required by the brief.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: An attacker or recording failure suppresses or alters source-required audit evidence so that access decisions, security events or admin actions cannot be reconstructed.

Mitigation: Compare source-required events with the resulting audit evidence; assess missing and altered-record cases and confirm unprovided protection or retention decisions. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-006; FR-007; FR-008; FR-009

Evidence ids: E-003

Owner: TBD

Id: RISK-005

Description: Derived risk affecting Monitoring and security alerts: Relevant suspicious activity could remain undetected or receive delayed investigation.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: Derived hypothesis: Suspicious activity evades the source-defined detection workflow, or excessive events obscure actionable alerts before an authorized reviewer can investigate.

Mitigation: Evaluate the stated detection duties with representative positive and negative events; confirm undecided thresholds, alert priorities and investigation responsibilities. Assign a responsible reviewer. Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-006; FR-007; FR-008; FR-009

Evidence ids: E-003

Owner: TBD

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Authorized network administrators

Goal: The system shall allow Authorized network administrators to review current configurations.

Story: As Authorized network administrators, I need the following behavior: The system shall allow Authorized network administrators to review current configurations.

Preconditions

1. Authorized network administrators review current configurations

Steps

1. initiate the review of current configurations

Success outcome: the requested configuration items are returned for examination

Requirement ids: FR-002

Evidence ids: E-002

Id: UC-002

Actor: Authorized network administrators

Goal: The system shall allow Authorized network administrators to submit configuration changes.

Story: As Authorized network administrators, I need the following behavior: The system shall allow Authorized network administrators to submit configuration changes.

Preconditions

1. Authorized network administrators submit a configuration change

Steps

1. initiate submission of a configuration change on the named object

Success outcome: the object's resulting policy, job, submitted change, or history is observed as updated

Requirement ids: FR-003

Evidence ids: E-002

Id: UC-003

Actor: Authorized network administrators

Goal: The system shall allow Authorized network administrators to track change history.

Story: As Authorized network administrators, I need the following behavior: The system shall allow Authorized network administrators to track change history.

Preconditions

1. Authorized network administrators track change history

Steps

1. initiate tracking of change history on the named object

Success outcome: the object's resulting policy, job, submitted change, or history is observed as accessible and accurate

Requirement ids: FR-004

Evidence ids: E-002

Id: UC-004

Actor: Authorized network administrators

Goal: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.

Story: As Authorized network administrators, I need the following behavior: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.

Preconditions

1. Authorized network administrators restore previously approved configurations

Steps

1. initiate restoration of a previously approved configuration on the named object

Success outcome: the object's resulting policy, job, submitted change, or restored state is observed as valid and consistent with approval records

Requirement ids: FR-005

Evidence ids: E-002

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: the system manages configuration changes across routers, switches, wireless controllers, and other network infrastructure
2. When: the system initiates a configuration change operation

Expected result: Expected: the resulting configuration is applied without error to the targeted device

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-002

Requirement ids: FR-002

Procedure

1. Given: Authorized network administrators review current configurations
2. When: initiate the review of current configurations

Expected result: Expected: the requested configuration items are returned for examination

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-003

Requirement ids: FR-003

Procedure

1. Given: Authorized network administrators submit a configuration change
2. When: initiate submission of a configuration change on the named object

Expected result: Expected: the object's resulting policy, job, submitted change, or history is observed as updated

Evidence to record: Evidence: External test capture of the resulting state

Test type: analysis

Id: TEST-004

Requirement ids: FR-004

Procedure

1. Given: Authorized network administrators track change history
2. When: initiate tracking of change history on the named object

Expected result: Expected: the object's resulting policy, job, submitted change, or history is observed as accessible and accurate

Evidence to record: Evidence: External test capture of the inspection record

Test type: positive

Id: TEST-005

Requirement ids: FR-005

Procedure

1. Given: Authorized network administrators restore previously approved configurations
2. When: initiate restoration of a previously approved configuration on the named object

Expected result: Expected: the object's resulting policy, job, submitted change, or restored state is observed as valid and consistent with approval records

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-006

Requirement ids: FR-006

Procedure

1. Given: the system detects an unauthorized or unexpected configuration change
2. When: the system initiates the detection process

Expected result: Expected: an alert is generated and logged in the audit trail

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-007

Requirement ids: FR-007

Procedure

1. Given: a security alert condition is triggered by an anomalous configuration event
2. When: the system evaluates and responds to the alert condition

Expected result: Expected: the alert exists in the designated monitoring channel as captured externally

Evidence to record: Evidence: External test capture of the resulting state

Test type: inspection

Id: TEST-008

Requirement ids: FR-008

Procedure

1. Given: an administrator modifies a network configuration
2. When: the system records the action and all associated metadata

Expected result: Expected: The system records the action and all associated metadata.

Evidence to record: Evidence: External test capture of the inspection record

Test type: analysis

Id: TEST-009

Requirement ids: FR-009

Procedure

1. Given: an administrator attempts to perform a restricted administrative function
2. When: the system evaluates the role-based access policy

Expected result: Expected: the request is denied if the role lacks approved permissions

Evidence to record: Evidence: External test capture of the resulting state

Test type: negative

Id: TEST-010

Requirement ids: FR-010

Procedure

1. Given: system provides a centralized dashboard
2. When: the system initiates the display of network configuration data

Expected result: Expected: The system initiates the display of network configuration data and the external test capture records the resulting response.

Evidence to record: Evidence: External test capture of the resulting response

Test type: positive

Id: TEST-011

Requirement ids: FR-011

Procedure

1. Given: system integrates with existing university identity services
2. When: the system performs a login authentication request via university identity protocols

Expected result: Expected: an authenticated session is established and persisted across operations

Evidence to record: Evidence: External test capture of the resulting state

Test type: positive

Id: TEST-012

Requirement ids: FR-012

Procedure

1. Given: system supports multiple campus locations
2. When: the system is initiated from each stated campus location

Expected result: Expected: use succeeds at all locations with consistent configuration access

Evidence to record: Evidence: External test capture of the inspection record

Test type: positive

Id: TEST-013

Requirement ids: NFR-001

Procedure

1. Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data exfiltration via network interfaces.
2. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD.

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or response metrics are stated in the draft plan.

Evidence to record: Evidence: External test capture of the inspection record

Test type: inspection

Id: TEST-014

Requirement ids: NFR-002

Procedure

1. Given: The system shall remain secure under all operational conditions.
2. When: An analyst reviews the security assurance plan and evaluates whether it includes measurable targets for threat detection, response time, and configuration drift

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-015

Requirement ids: NFR-003

Procedure

1. Given: The system shall remain reliable under all operational conditions.
2. When: An analyst reviews the reliability assurance plan and evaluates whether it includes measurable targets for failure detection, recovery time, and service uptime

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-016

Requirement ids: NFR-004

Procedure

1. Given: The system shall remain scalable under all operational conditions.
2. When: An analyst reviews the scalability assurance plan and evaluates whether it includes measurable targets for throughput, latency, and resource utilization growth

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Id: TEST-017

Requirement ids: NFR-005

Procedure

1. Given: The system shall remain manageable under all operational conditions.
2. When: An analyst reviews the manageability assurance plan and evaluates whether it includes measurable targets for configuration change frequency, audit coverage, and

Expected result: Expected: The plan's measurable targets remain TBD or unconfirmed

Evidence to record: Evidence: External test capture of the resulting response

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-003

Log

Detail: A collection of event records. This term does not itself require additional logging fields, events or access permissions.

Status: proposed

Evidence ids: E-003

Access restoration

Detail: Returning access that was previously restricted, within the source-defined permission boundary.

Status: proposed

Evidence ids: E-002

14. SDLC recommendation and workflow

Selected model: Incremental

Justification: Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003) Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004) Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.
2. Proposed suitability recommendation; confirm delivery constraints and team capacity before adopting this lifecycle. Missing factors are not assumed.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure.; FR-002: The system shall allow Authorized network administrators to review current configurations.; FR-003: The system shall allow Authorized network administrators to submit configuration changes.; FR-004: The system shall allow Authorized network administrators to track change history.; FR-005: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.; FR-006: The system shall detect unauthorized or unexpected configuration changes.; FR-007: The system shall generate security alerts.; FR-008: The system shall maintain audit logs of configuration activity and administrator actions.; FR-009: The system shall restrict administrative functions based on approved roles.; FR-010: The system shall provide a centralized dashboard.; FR-011: The system shall integrate with existing university identity services.; FR-012: The system shall support multiple campus locations.
2. Review and baseline the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-007: The system shall generate security alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Incremental delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: The system shall manage configuration changes across routers, switches, wireless controllers, and other network infrastructure.; FR-002: The system shall allow Authorized network administrators to review current configurations.; FR-003: The system shall allow Authorized network administrators to submit configuration changes.; FR-004: The system shall allow Authorized network administrators to track change history.; FR-005: The system shall allow Authorized network administrators to restore previously approved configurations when necessary.; FR-006: The system shall detect unauthorized or unexpected configuration changes.; FR-007: The system shall generate security alerts.; FR-008: The system shall maintain audit logs of configuration activity and administrator actions.; FR-009: The system shall restrict administrative functions based on approved roles.; FR-010: The system shall provide a centralized dashboard.; FR-011: The system shall integrate with existing university identity services.; FR-012: The system shall support multiple campus locations.
3. Implement the source-defined work group: NFR-001: The system shall be secure.; NFR-002: The system shall remain secure.; NFR-003: The system shall remain reliable.; NFR-004: The system shall remain scalable.; NFR-005: The system shall remain manageable.
4. In an authorized isolated test environment, execute TEST-001: Given: the system manages configuration changes across routers, switches, wireless controllers, and other network infrastructure When: the system initiates a configuration change operation Expected: the resulting configuration is applied without error to the targeted device
5. In an authorized isolated test environment, execute TEST-002: Given: Authorized network administrators review current configurations When: initiate the review of current configurations Expected: the requested configuration items are returned for examination
6. In an authorized isolated test environment, execute TEST-003: Given: Authorized network administrators submit a configuration change When: initiate submission of a configuration change on the named object Expected: the object's resulting policy, job, submitted change, or history is observed as updated
7. In an authorized isolated test environment, execute TEST-004: Given: Authorized network administrators track change history When: initiate tracking of change history on the named object Expected: the object's resulting policy, job, submitted change, or history is observed as accessible and accurate

8. In an authorized isolated test environment, execute TEST-005: Given: Authorized network administrators restore previously approved configurations When: initiate restoration of a previously approved configuration on the named object Expected: the object's resulting policy, job, submitted change, or restored state is observed as valid and consistent with approval records
9. In an authorized isolated test environment, execute TEST-006: Given: the system detects an unauthorized or unexpected configuration change When: the system initiates the detection process Expected: an alert is generated and logged in the audit trail
10. In an authorized isolated test environment, execute TEST-007: Given: a security alert condition is triggered by an anomalous configuration event When: the system evaluates and responds to the alert condition Expected: the alert exists in the designated monitoring channel as captured externally
11. In an authorized isolated test environment, execute TEST-008: Given: an administrator modifies a network configuration When: the system records the action and all associated metadata Expected: The system records the action and all associated metadata.
12. In an authorized isolated test environment, execute TEST-009: Given: an administrator attempts to perform a restricted administrative function When: the system evaluates the role-based access policy Expected: the request is denied if the role lacks approved permissions
13. In an authorized isolated test environment, execute TEST-010: Given: system provides a centralized dashboard When: the system initiates the display of network configuration data Expected: The system initiates the display of network configuration data and the external test capture records the resulting response.
14. In an authorized isolated test environment, execute TEST-011: Given: system integrates with existing university identity services When: the system performs a login authentication request via university identity protocols Expected: an authenticated session is established and persisted across operations
15. In an authorized isolated test environment, execute TEST-012: Given: system supports multiple campus locations When: the system is initiated from each stated campus location Expected: use succeeds at all locations with consistent configuration access
16. In an authorized isolated test environment, execute TEST-013: Given: A draft acceptance plan for this quality goal: the system shall be secure against unauthorized access and data exfiltration via network interfaces. When: An analyst inspects or reviews that acceptance plan to determine whether measurable targets are defined or remain TBD. Expected: The plan's measurable targets remain TBD or unconfirmed, as no specific thresholds, attack scenarios, or response metrics are stated in the draft plan.
17. In an authorized isolated test environment, execute TEST-014: Given: The system shall remain secure under all operational conditions. When: An analyst reviews the security assurance plan and evaluates whether it includes measurable targets for threat detection, response time, and configuration drift Expected: The plan's measurable targets remain TBD or unconfirmed
18. In an authorized isolated test environment, execute TEST-015: Given: The system shall remain reliable under all operational conditions. When: An analyst reviews the reliability assurance plan and evaluates whether it includes measurable targets for failure detection, recovery time, and service uptime Expected: The plan's measurable targets remain TBD or unconfirmed
19. In an authorized isolated test environment, execute TEST-016: Given: The system shall remain scalable under all operational conditions. When: An analyst reviews the scalability assurance plan and evaluates whether it includes measurable targets for throughput, latency, and resource utilization growth Expected: The plan's measurable targets remain TBD or unconfirmed
20. In an authorized isolated test environment, execute TEST-017: Given: The system shall remain manageable under all operational conditions. When: An analyst reviews the manageability assurance plan and evaluates whether it includes measurable targets for configuration change

frequency, audit coverage, and Expected: The plan's measurable targets remain TBD orunconfirmed

21. Collect stakeholder feedback; agree its cadence with a human reviewer before scheduling cycles.
22. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-007: The system shall generate security alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-007: The system shall generate security alerts.
2. Review implementation and acceptance evidence against NFR-001: The system shall be secure.
3. Review implementation and acceptance evidence against NFR-002: The system shall remain secure.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; FR-004; FR-005; FR-006; FR-007; FR-008; FR-009; FR-010; FR-011; FR-012; NFR-001; NFR-002; NFR-003; NFR-004; NFR-005

Unknown factors: budget; delivery; failure_consequences; feedback; legacy; regulation; schedule; stability; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-2.0

Generation profile: sequential-sdlc-generation-2.1

Factors

Name: security_assurance

Value: integrated security review

Evidence ids: E-002

Basis: derived

Rationale: Security decisions benefit from security review and tests throughout delivery.

Name: operational_validation

Value: monitoring feedback

Evidence ids: E-003

Basis: derived

Rationale: Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency.

Name: deployment_scope

Value: staged rollout opportunity

Evidence ids: E-004

Basis: derived

Rationale: Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed.

Name: integration_assurance

Value: integration validation

Evidence ids: E-004

Basis: derived

Rationale: Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified.

Ranked candidates

Model: Incremental

Score: 66

Reasons: Derived operational_validation (+6 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+12 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004); Derived integration_assurance (+8 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Hybrid

Score: 65

Reasons: Derived security_assurance (+8 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+5 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+8 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004); Derived integration_assurance (+4 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: DevSecOps

Score: 63

Reasons: Derived security_assurance (+12 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived operational_validation (+8 points): Monitoring and alert behavior needs validation with operational feedback; this does not establish continuous deployment or a release frequency. (E-003); Derived deployment_scope (+3 points): Multiple deployment locations offer a staged validation and rollout opportunity; the rollout plan remains proposed. (E-004)

Model: Spiral

Score: 51

Reasons: Derived security_assurance (+5 points): Security decisions benefit from security review and tests throughout delivery. (E-002); Derived integration_assurance (+6 points): Connected systems need integration verification before broader rollout; their age, technology and technical uncertainty remain unspecified. (E-004)

Model: Agile

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 40

Reasons: No differentiating input evidence

Evidence register

Id: E-001

Source: description

Quote: Design a secure network configuration management system for a university that manages configuration changes across routers, switches, wireless controllers, and other network infrastructure.

Id: E-002

Source: description

Quote: Authorized network administrators should be able to review current configurations, submit configuration changes, track change history, and restore previously approved configurations when necessary.

Id: E-003

Source: description

Quote: The system should detect unauthorized or unexpected configuration changes, generate security alerts, maintain audit logs of configuration activity and administrator actions, and restrict administrative functions based on approved roles.

Id: E-004

Source: description

Quote: The platform should provide a centralized dashboard, integrate with existing university identity services, support multiple campus locations, and remain secure, reliable, scalable, and manageable.

Source references

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.7216

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7041

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-f68d66ab02ee9ba0:14 - 03_OWASP_CheatSheets/073_OWASP_Logging_Cheat_Sheet.md

Source document id: kb-f68d66ab02ee9ba0

Chunk index: 14

Page or section: text chunk 15

Relevance score: 0.6858

Sha256: e3965a340b575bd232c7f5c7bf24d96a4e22bace1783a7a3a11a84d409e0b560

kb-f68d66ab02ee9ba0:13 - 03_OWASP_CheatSheets/073_OWASP_Logging_Cheat_Sheet.md

Source document id: kb-f68d66ab02ee9ba0

Chunk index: 13

Page or section: text chunk 14

Relevance score: 0.694

Sha256: e3965a340b575bd232c7f5c7bf24d96a4e22bace1783a7a3a11a84d409e0b560

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.6952

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.6825

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.35

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-001	TEST-001, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-002	E-002	UC-001, TEST-002, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-003	E-002	UC-002, TEST-003, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-004	E-002	UC-003, TEST-004, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-005	E-002	UC-004, TEST-005, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-006	E-003	TEST-006, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-007	E-003	TEST-007, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-008	E-003	TEST-008, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-009	E-003	TEST-009, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-010	E-004	TEST-010, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-011	E-004	TEST-011, Requirements and verification planning, Incremental delivery cycle, Release and handoff
FR-012	E-004	TEST-012, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-001	E-001	TEST-013, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-002	E-004	TEST-014, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Requirement	Evidence / sources	Stories / tests / SDLC
NFR-003	E-004	TEST-015, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-004	E-004	TEST-016, Requirements and verification planning, Incremental delivery cycle, Release and handoff
NFR-005	E-004	TEST-017, Requirements and verification planning, Incremental delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.35